

BStick: Hand-held Haptic Controller for Virtual Reality Applications

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ABSTRACT

This study proposes Bstick-Mark2, a handheld virtual reality (VR) haptic controller that monitors and controls input from five fingers in real-time with pressure sensors and linear motors. Bstick-Mark2 is designed and fabricated to enable users to use both hands along with a head-mounted display (HMD) while freely roaming based on Bluetooth technologies. When a user holds the haptic controller with fingers, the data input from five pressure sensors is transmitted to microcontroller unit (MCU) to independently control the movements of five linear motors. The pressure and position data of linear motors are sent to a computer connected to a VR display through a Bluetooth module embedded in the controller and utilized in interaction with a virtual object and virtual hand movements using the Unity game engine. Bstick-Mark2 can withstand 22 N of force per finger to maintain the pressing force of a male's finger and is compact to enable users to easily handle using their hands. It enables to make sensations of grabbing and controlling while interacting with VR content.

CCS CONCEPTS

• **Hardware** → **Emerging interfaces.**

KEYWORDS

handheld controller, haptic device, virtual reality

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1 INTRODUCTION

Recent advancements in virtual reality (VR) technology have enabled users to experience a high-quality virtual space through

stereoscopic display and spatial audio technology, particularly requiring various mechanisms of using hands for complex interactions. For commercial VR controllers, users control virtual objects with buttons, a joystick, and a touchpad of a controller with both hands; however, the finger position for operating the controller remains constant even if the shape of virtual objects changes. Precise haptic feedback can be provided using an exoskeleton device; however, a handheld controller is more appropriate considering the usability and convenience of wearing a controller when a user is already wearing a VR display device [Aiple and Schiele 2013; Gu et al. 2016]. However, a handheld controller must have all mechanical and electronic mechanisms in a limited size for hand-holding. Accordingly, research is being conducted on controllers with special mechanisms that enable a device to provide feedback appropriate for specific purposes [Choi et al. 2017, 2018; Lee et al. 2019]. To overcome the above-mentioned limitations, this study examined a handheld device installed with a linear actuator generating force sensation in all five fingers. Considering the actuator embedded in the controller can withstand the pressing force of an adult finger, it can render hard virtual objects. Furthermore, soft objects can also be rendered by controlling the movements of actuators when the pressing force of a user's finger is monitored. This study miniaturized the hardware to enable users to easily wear Bstick-Mark2, a handheld VR controller equipped with five linear motors, which enables length adjustments using fingers and advanced software to create a user experience where virtual objects can be grabbed and controlled in a VR environment.

2 HANDHELD HAPTIC CONTROLLER

As shown in Figure 1, Bstick-Mark1 and Mark2 are designed as a handheld device and includes a driving part as buttons that come in contact with five fingers. A force sensor attached to each finger module measures the pressure applied when a user presses a button. A trackpad is additionally installed in the thumb part to measure the contact location, and the trackpad brings up a user interface with the properties of a button when the trackpad is pressed [Ko and Nam 2022].

Bstick consists of three modules. The first module is a force feedback module, which includes five micro linear actuators. This module comprises a driving module that drives the micro linear actuators and collects data from the sensors of inside the device, a network module for exchanging data with a PC, and a main

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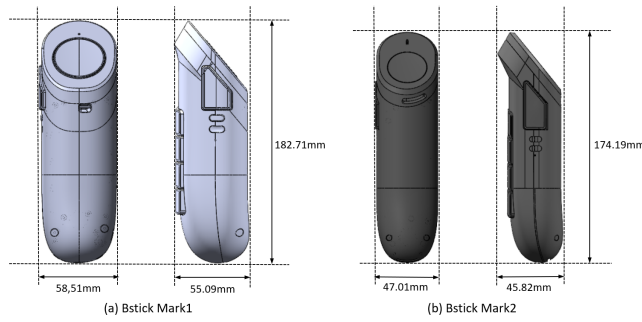


Figure 1: Exterior design of Bstick.

PCB module with a control module that controls the controller. The second module, a grip & vibration module, includes an external case for the controller where a battery is arranged on the back of the case, and a vibrator also arranged on the backside generates vibration; this module also includes a USB PCB to enable USB communications. The final module, a touch button module, is arranged at the top of the controller.

Bstick-Mark2, redesigned to address the external design and inter-device interference issues of Bstick-Mark1, integrates five linear actuators and includes a micro-DC geared motor with a gear ratio of 30:1 and a linear potentiometer developed by the research team as shown in Figure 2. Four PCBs used in Bstick-Mark1 were reduced to three in Bstick-Mark2, with each PCB connected to a flexible printed circuit (FPC) separately designed to minimize the overall size and enhance the level of mechanical completion as shown in Figure 3.

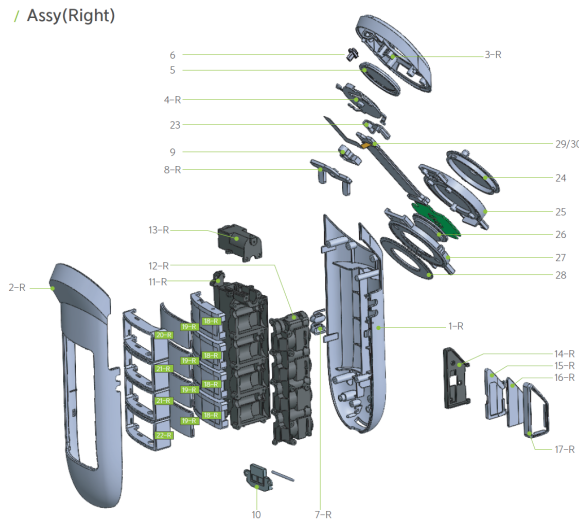


Figure 2: Design of Bstick-Mark2's Internal Module.

3 CONCLUSION

This study proposed Bstick-Mark2 is a haptic controller capable of independently controlling five fingers. Considering Bstick can



Figure 3: Implemented Bstick-Mark2.

control button position using linear actuators for each of the five fingers, objects can be rendered in a virtual space with adjustable stiffness and ductility, allowing for precise grasping and manipulating. Linear actuators were miniaturized and integrated to reduce the size of the grip of Bstick-Mark1, which has been previously researched, for improving the user interface; in addition, the exterior size of the haptic feedback module and controller was also reduced to fabricate Bstick-Mark2. Compared to Bstick-Mark1, the intensity of force feedback was improved by 7.1–9.63 N, and the moving stance of the force feedback module was improved by 0.60–1.08 mm, thus improving the general performance of the haptic feedback module.

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REFERENCES

- Manuel Aiple and André Schiele. 2013. Pushing the limits of the CyberGrasp™ for haptic rendering. In *2013 IEEE international conference on robotics and automation*. IEEE, 3541–3546.
- Inrak Choi, Heather Culbertson, Mark R Miller, Alex Olwal, and Sean Follmer. 2017. Grabity: A wearable haptic interface for simulating weight and grasping in virtual reality. In *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology*. 119–130.
- Inrak Choi, Eyal Ofek, Hrvoje Benko, Mike Sinclair, and Christian Holz. 2018. Claw: A multifunctional handheld haptic controller for grasping, touching, and triggering in virtual reality. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. 1–13.
- Xiaochi Gu, Yifei Zhang, Weize Sun, Yuanzhe Bian, Dao Zhou, and Per Ola Kristensson. 2016. Dexmo: An inexpensive and lightweight mechanical exoskeleton for motion capture and force feedback in VR. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. 1991–1995.
- Ginam Ko and SangHun Nam. 2022. Bstick: Handheld Virtual Reality Haptic Controller for Hand Rehabilitation. *Systems* 10, 3 (2022), 54.
- Jayeon Lee, Mike Sinclair, Mar Gonzalez-Franco, Eyal Ofek, and Christian Holz. 2019. TORC: A virtual reality controller for in-hand high-dexterity finger interaction. In *Proceedings of the 2019 CHI conference on human factors in computing systems*. 1–13.